Nicholas Lee

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LinkedIn
Portfolio

Skills

- **Mechanical Design:** SolidWorks, AutoCAD, OnShape, FEA, DFM, DFA, GD&T, Surface Modeling
- **Manufacturing:** 3D Printing, Milling, Sheet Metal, Root Cause Analysis, Preventative Maintenance, BOM
- **Programming:** C++, Python, Simulink/MATLAB, RTOS, Git, Ladder Logic (Siemens TIA Portal)

Experience

Mechanical Designer – Midnight Sun Solar Car Team – Waterloo, ON Sept 2025 – Present

- Reduced component weight by 20% while maintaining structural integrity by designing an aluminum sheet metal ballast box in SolidWorks using DFM/A principles and static FEA
- Improved wheel cover assembly accuracy and reduced installation errors by modeling wheel cover cutting stencils referenced to vehicle datum features
- Accelerated component lead time from weeks to hours, by designing and prototyping press-fit camera mount system for 3D printing

Undergraduate Research Assistant – MSAM Lab – Waterloo, ON Jan 2026 – Present

- Decreased gear design cycle time through the development of parametric CAD models in SolidWorks
- Ensured reliable operation of educational gearbox assemblies by iteratively designing components in SolidWorks optimized for FDM 3D Printing
- Streamlined the effectiveness of hands-on educational demonstrations by integrating DC motors into gearbox systems to enable automated actuation

Mechanical Team Member – Waterloo Automation Collective – Waterloo, ON Jan 2026 – Present

- Accelerated design iteration speed and centralized CAD management by developing parametrically driven clamping linkages in OnShape
- Pioneered the development of the university's first desktop injection molder by driving early-stage design and prototyping of key subsystems as a founding member

Engineering Intern – Refresco Beverages – Mississauga, ON Jan 2025 – Aug 2025

- Reduced bottle fill errors by 50% by performing a root cause analysis on production datasets in Excel to identify and mitigate filling level variances
- Extended expected service life of a pneumatic piston assembly by 200% by reverse-engineering assembly components in SolidWorks to mitigate sugar buildup
- Reduced unplanned downtime through the creation of comprehensive Technical Documentation (task lists and drawings) and BOMs for integration into the CMMS
- Minimized installation uncertainty and resolved spatial conflicts for a production line reformat by drafting precise platform locations in AutoCAD

Projects

Arduino Path Following RoboCar

- Programmed an autonomous line-following robot in C++ (Arduino IDE), achieving stable navigation through a figure-eight path and inclined ramp using light sensor distance logic

Education

University of Waterloo - BASc, Mechatronics Engineering, GPA: 3.85. Dean's Honours Sept 2023 – May 2028
Coursework: Mechanics of Deformable Solids, Thermodynamics and Heat Transfer, Design and Dynamics of Machines

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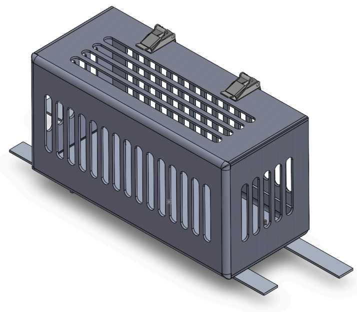
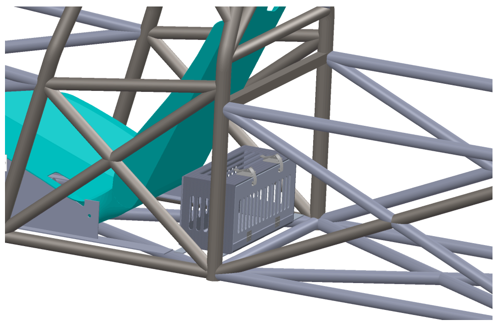
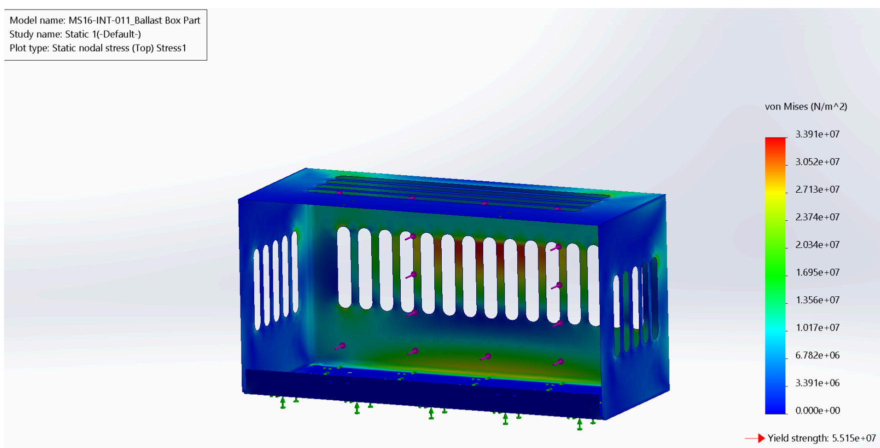
Ballast Box | SolidWorks, Sheet Metal, FEA, DFM/A

Designed an aluminum sheet metal ballast box in SolidWorks to hold a minimum 20 kg of steel shot. Performed static FEA to optimize geometry and achieve a 20% weight reduction while maintaining structural integrity under driving load cases (e.g., emergency braking impact loads). The design was validated for manufacturing and is currently being integrated into the competition vehicle chassis.

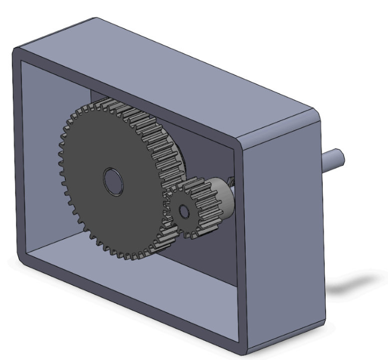

SolidWorks Model and Manufactured Assembly



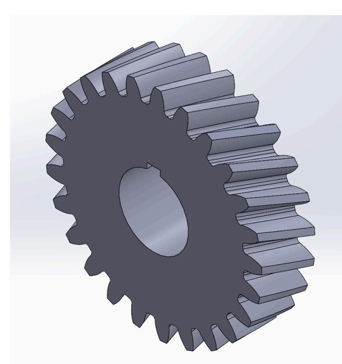
FEA Results

Two-Stage Gearboxes | 3D Printing, DC Motors, Gear Ratios, nTop

Designed and prototyped multiple two-stage gearboxes for undergraduate lectures. Introduced metal additive manufacturing techniques into the gearbox housing using nTop. Integrated DC motors to drive gear actuation for live demonstration during lectures. Parametrically modeled helix gear for easy design iterations.



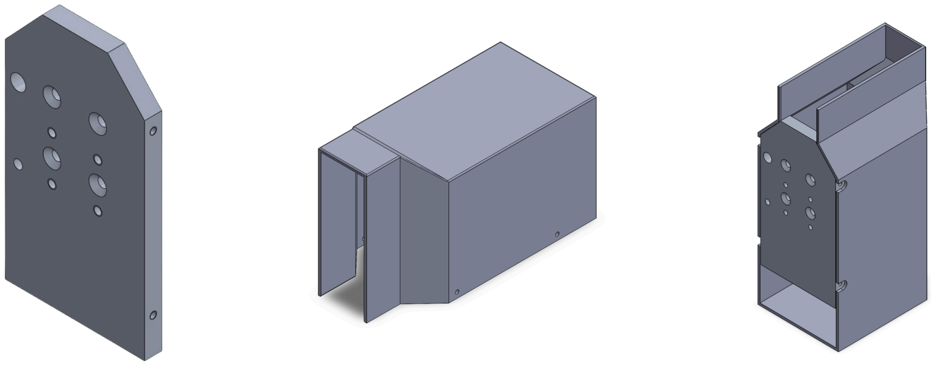
SolidWorks Model of Gearbox



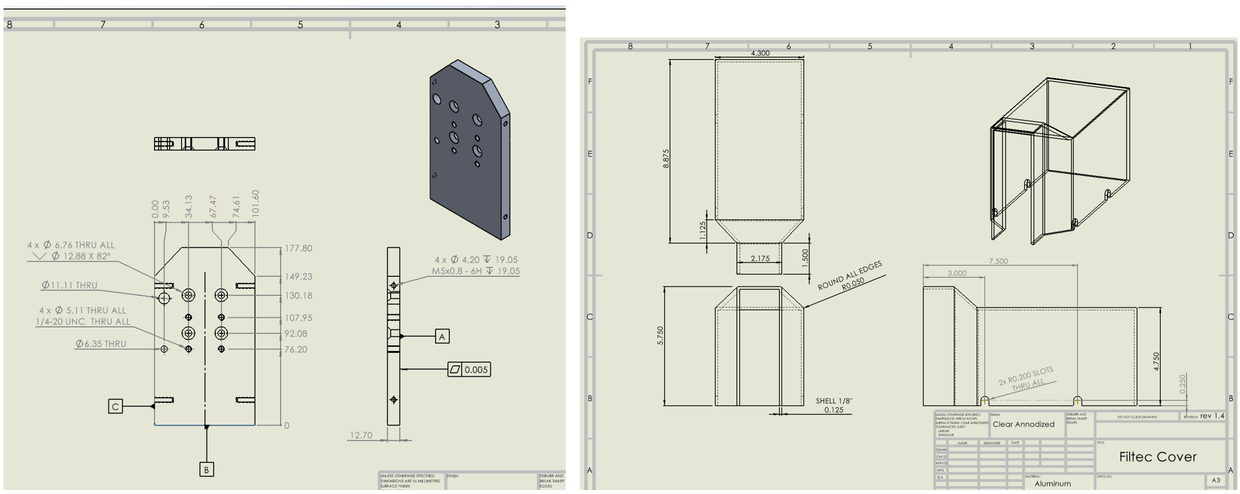
Parametric Gear Model

Pneumatic Piston Mounting Assembly | Technical Drawings, Reverse-Engineering

Redesigned a pneumatic piston assembly to address syrup buildup causing mistimed can kickouts. Adapted the mounting plate to integrate a replacement piston from another vendor . Designed a protective cover to prevent syrup splashes. Increased service life by 200%, significantly reducing maintenance frequency and downtime.



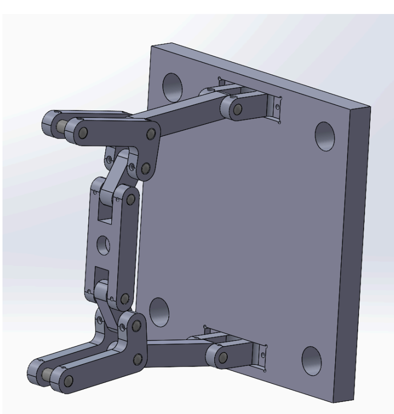
SolidWorks Model

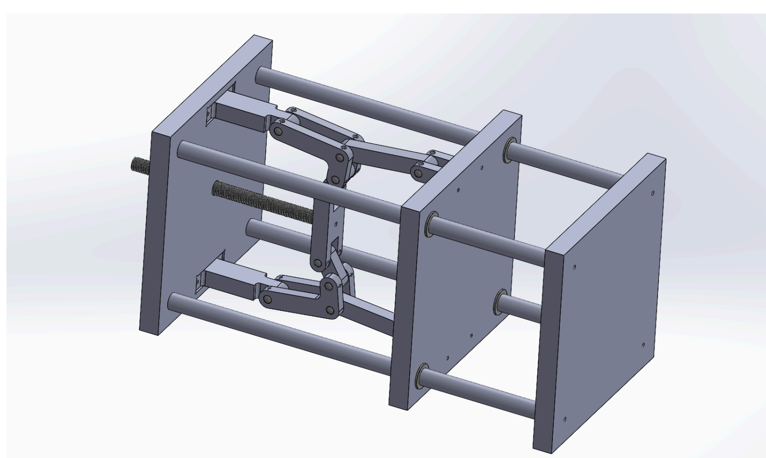


Technical Drawings

WAC-A-MOLD | Parametric Design, Machining

One of the core founding members of the Waterloo Automation Collective, building a desktop injection molder. Parametrized the clamping unit linkages through SolidWorks to ensure easy design iterations. Machining the links using a mill.





Linkages **Full Clamping Unit**